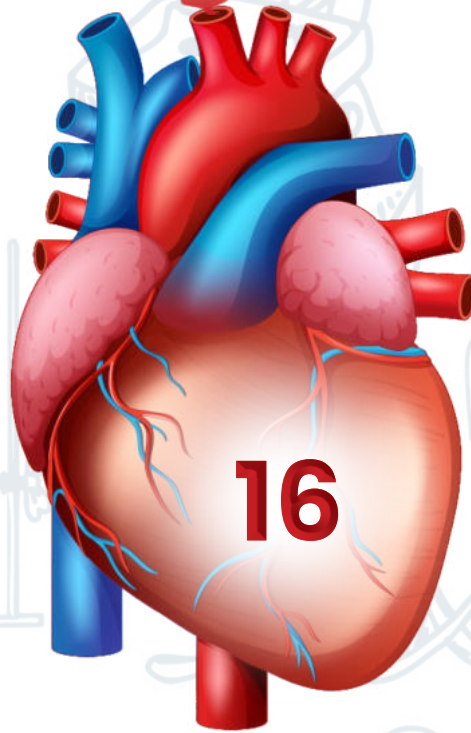
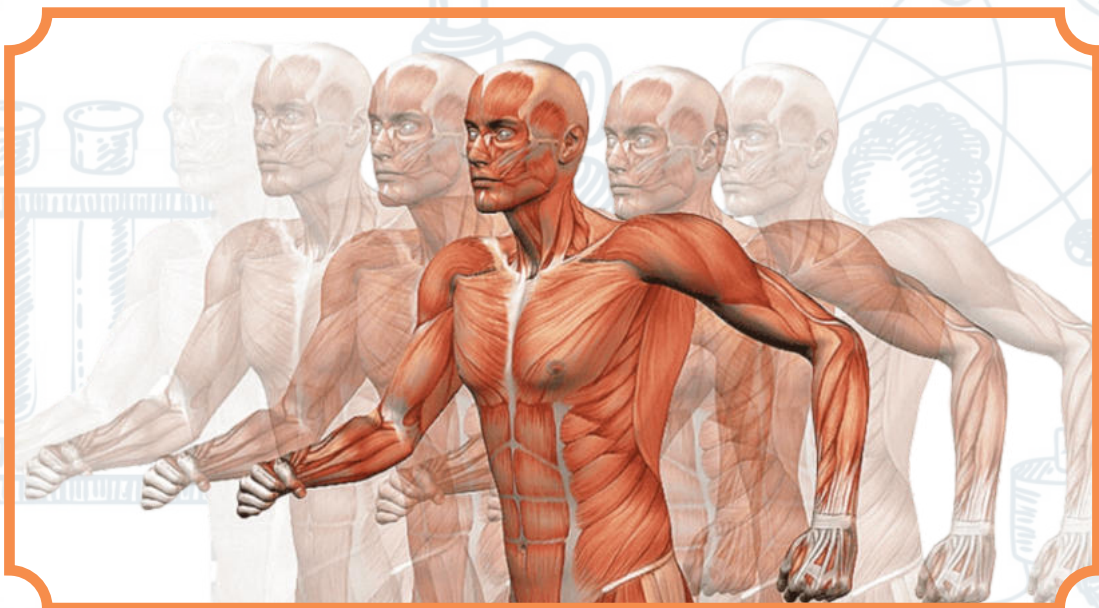


UNIT



SUPPORT & MOVEMENT





Human Skeleton

1. Definition

- Endoskeleton: Internal support structure of the body; lies within soft tissues.

2. Components

- Bones.
- Cartilage.

3. Major Functions

1. Support & Shape

- Provides rigid framework.

2. Protection

- Shields vital organs (e.g., cranium protects brain, rib cage protects heart & lungs).

3. Hematopoiesis

- Red bone marrow produces red & white blood cells, platelets.

4. Mineral Storage

- Reservoir for calcium, phosphorus, and other minerals.

4. Study of Bones

- **Osteology:** Branch of anatomy/science dedicated to the study of bones.

5. Musculoskeletal Integration

● Musculoskeletal System:

- Bones + muscles + connective tissues working together.
- Enables coordinated movement, posture, and balance.

6. Bone Count & Development

- At Birth: ~300 bones.
- In Adulthood: 206 bones.
- Reduction due to fusion (e.g., sacrum, coccyx).





1. Composition of Bone

- Cells (20% of organic matrix).
- Osteoblasts (bone-forming).
- Osteocytes (mature maintenance cells).
- Osteoclasts (bone-resorbing).
- Collagen Fibers (organic matrix).
- Type I collagen provides tensile strength.
- Serves as scaffold for mineral deposition.
- Minerals (primarily hydroxyapatite).
- Calcium phosphate crystals confer hardness.
- Water (10–20%).

2. Bone Development Processes

- Calcification (Ossification).
- Deposition of calcium salts onto collagen fibers.
- Remodeling.
- Continuous balance of osteoblast and osteoclast activity.

3. Regions of a Long Bone

1. Epiphysis (terminal ends)

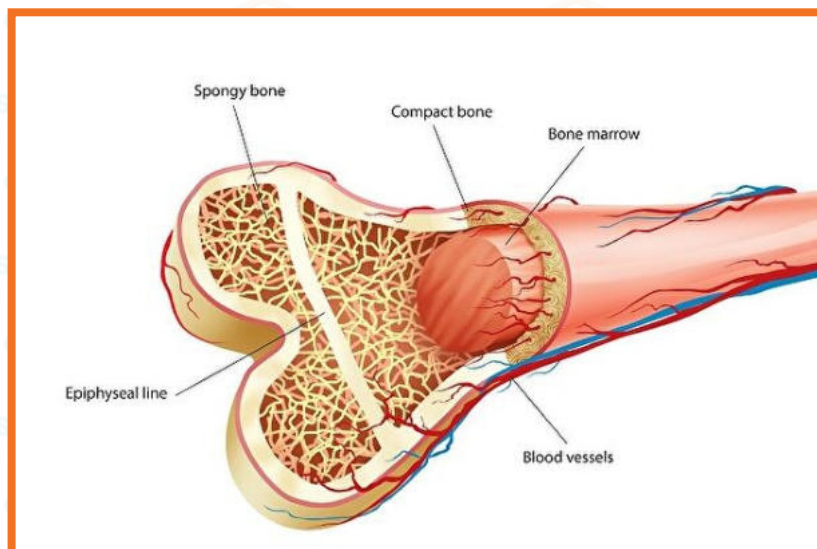
- Composed of cancellous (spongy) bone.
- Trabeculae: interlacing rods/plates that bear weight.
- Filled with red bone marrow (≈ 20% of total marrow mass; major site of hematopoiesis).

2. Metaphysis (between epiphysis & diaphysis).

- Contains growth plate (in growing bone).
- Transition zone for nutrient/mineral exchange.

3. Diaphysis (shaft/middle region)

- Cortical (compact) bone (80% of skeletal mass).
- Organized into osteons (Haversian systems).
- Each osteon comprises concentric lamellae (collagen + hydroxyapatite).
- Central Haversian canal: blood vessels & nerve fibers.
- Medullary (marrow) cavity.
- Hollow, lined by endosteum.
- Contains yellow marrow (fat) in adults; red marrow in infants.





4. Microarchitecture

- Osteon (Haversian System).
- Lamellae: concentric rings of matrix.
- Lacunae: small spaces housing osteocytes.
- Canaliculi: tiny canals for intercellular fluid/nutrient exchange.
- Trabeculae (in spongy bone).
- Align along stress lines.
- Reduce bone weight while maintaining strength.

Bone Coverings & Cellular Components

1. Bone Coverings

- Periosteum (Outer Layer).
- Dense, fibrous membrane enveloping external bone surface (except at articular regions).
- Attachment site for tendons and ligaments.
- Rich in blood vessels & lymphatics—provides nourishment and waste removal.
- Contains osteogenic progenitor cells in its inner cambium layer.
- Endosteum (Inner Layer).
- Thin, vascular membrane lining the medullary cavity and Haversian canals.
- Houses osteogenic progenitor cells.
- Critical in bone growth, remodeling, and fracture repair.

2. Osteogenic Progenitor Cells

- Undifferentiated stem cells residing in periosteum and endosteum.
- **Differentiate into:**
- Osteoblasts (bone-matrix secreting).
- Chondroblasts (cartilage-matrix secreting).

3. Bone Cell Types & Functions

1. Osteoblasts:

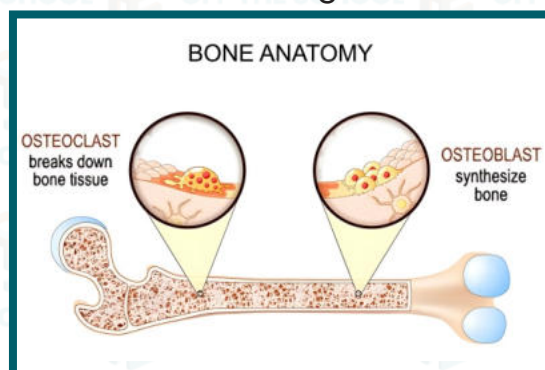
- Derive from osteogenic progenitors.
- Synthesize and secrete organic bone matrix (osteoid).
- Initiate mineralization; become entrapped in matrix.

2. Osteocytes:

- Mature osteoblasts residing within lacunae.
- Maintain bone matrix; sense mechanical stress.
- Direct remodeling by signaling to osteoblasts and osteoclasts.

3. Osteoclasts:

- Large, multinucleated cells originating from monocyte/macrophage lineage.
- Resorb bone by secreting acids and proteolytic enzymes,
- Essential for demineralization, remodeling, and calcium homeostasis,





4. Role in Fracture Healing:

- Periosteal and endosteal progenitors proliferate and differentiate.
- Osteoblasts deposit new matrix to form callus.
- Osteoclasts remodel callus into mature bone.

Structure of Cartilage

1. Definition & Location

- Cartilage: Firm, yet flexible connective tissue.
- Found in embryonic skeleton (precursor to bone) and adult joints, respiratory tract, ear, and intervertebral discs.

2. Covering & Nutrition

- Perichondrium: Dense, fibrous membrane enveloping cartilage.
- Contains blood vessels, nerves, and progenitor cells (chondroblasts).
- **Nutrition:**
- Avascular tissue—nutrients and oxygen diffuse through matrix from perichondrium or synovial fluid.
- Slow growth and repair due to lack of direct blood supply.

3. Cellular Components

1. Chondroblasts:

- Located in the inner layer of the perichondrium.
- Secrete cartilage matrix.
- Differentiate into chondrocytes once entrapped.

2. Chondrocytes:

- Mature cartilage cells residing in lacunae (small cavities).
- Maintain matrix and respond to mechanical stimuli.

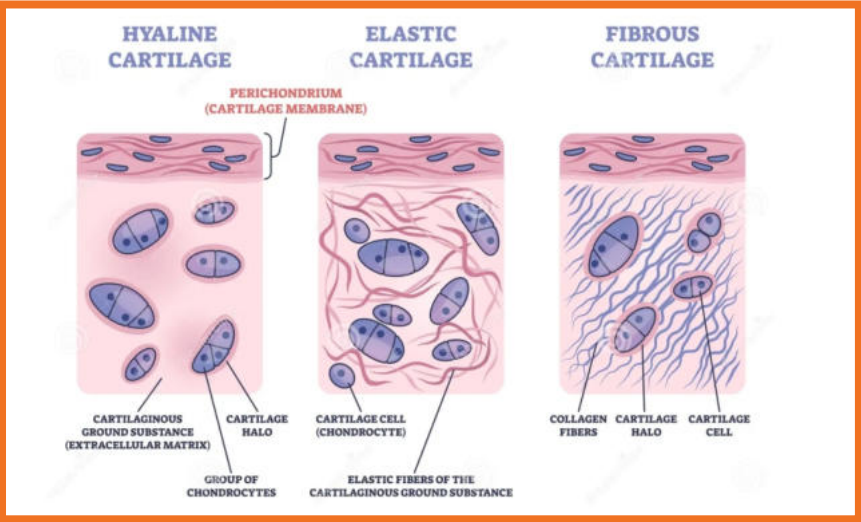
4. Extracellular Matrix

- Collagen Fibers.
- Primarily type III collagen for tensile strength.
- Proteoglycans.
- Core protein + glycosaminoglycans (e.g., chondroitin sulfate).
- Create a hydrated gel that resists compressive forces.
- Non-collagenous Proteins.
- Contribute to matrix organization and signaling.

5. Types of Cartilage

Type	Matrix Characteristics	Locations & Function
Hyaline	Smooth, glassy; fine collagen network	Articular surfaces of synovial joints; ribs–sternum articulations; nose; tracheal rings (facilitates smooth movement, cushioning)
Fibrocartilage	Dense, coarse collagen bundles	Intervertebral discs; pubic symphysis; menisci of knee (provides tensile strength and shock absorption)
Elastic	Abundant elastic fibers + collagen	External ear (pinna); epiglottis; auditory (Eustachian) tubes; larynx (offers flexibility with shape retention)

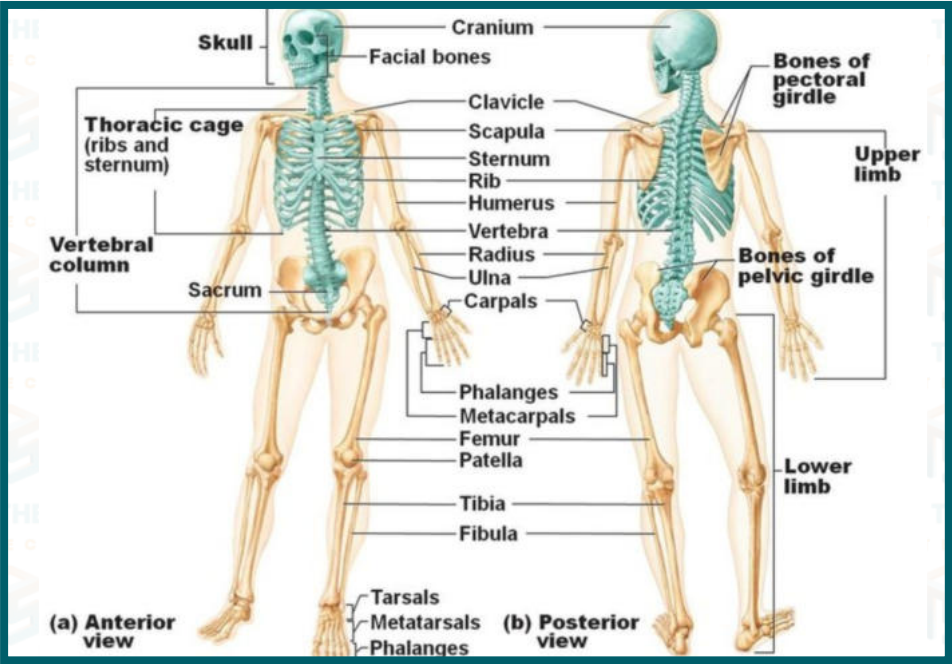




Divisions of the Human Skeleton

1. Overview:

- Total Bones (Adult): **206**
- Two Major Divisions:
 1. Axial Skeleton.
 2. Appendicular Skeleton.



2. Axial Skeleton:

1. Definition & Functions

- Location: Central "axis" of the body.
- **Primary Function:**
 - Protects vital organs (brain, spinal cord, thoracic viscera).
 - Provides attachment sites for muscles of head, neck, and trunk.
 - Supports weight of the head and trunk.
 - Participates in hematopoiesis (red marrow in vertebrae, ribs, sternum).

2. Bone Count

- Total Bones: **80**



Component	Number of Bones
Skull	28
Hyoid	1
Vertebral column	26
Ribs	24
Sternum	1
Total	80

3. Skull (28 Bones)

1. Cranial Bones (8)

- Frontal (1)
- Parietal (2)
- Temporal (2)
- Occipital (1)
- Sphenoid (1)
- Ethmoid (1)

2. Facial Bones (14)

Maxillae (2), Zygomatic (2), Nasal (2), Lacrimal (2), Palatine (2), Inferior nasal conchae (2), Vomer (1), Mandible (1).

Functions: protect facial organs; aid in breathing, eating, speech, and expressions.

3. Auditory Ossicles (6)

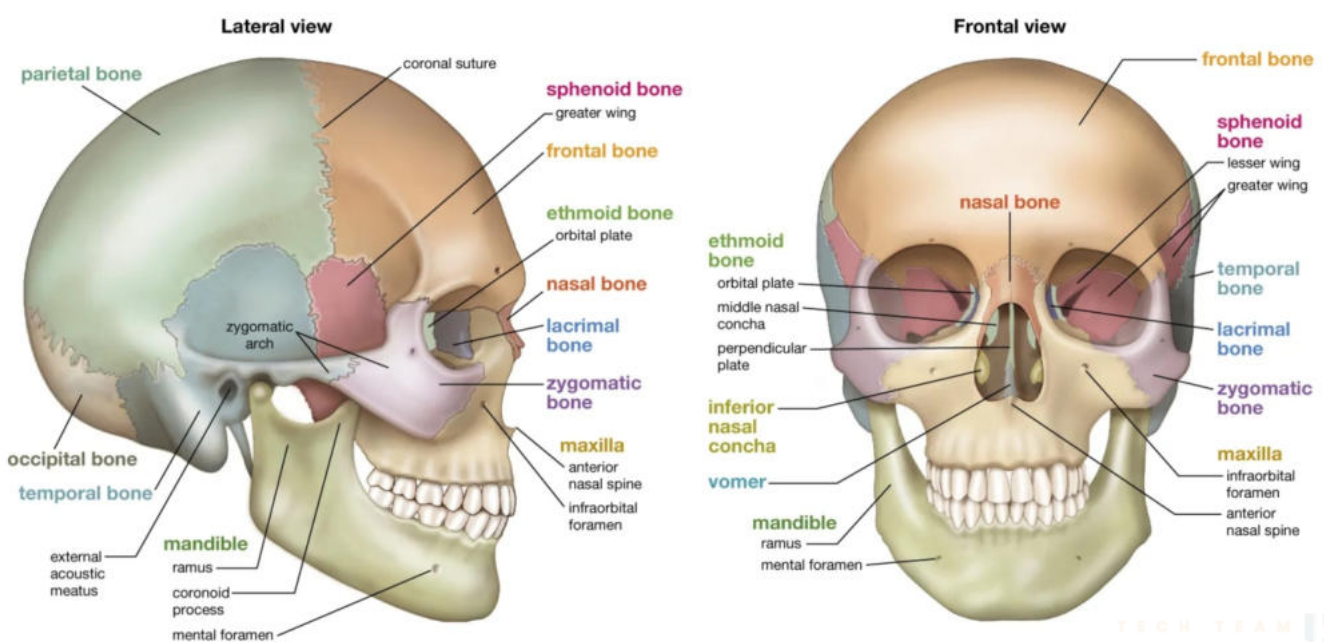
Malleus (2), Incus (2), Stapes (2).

Function: transmit sound vibrations from eardrum to inner ear.

4. Hyoid Bone (1)

- Shape: "U-shaped".
- Location: Between mandible and larynx, not articulated with other bones.

Function: Anchor for tongue and floor-of-mouth muscles.





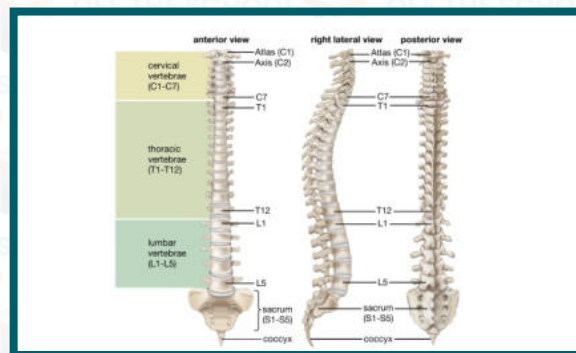
5. Vertebral Column (26 Bones)

Region	Count	Notes
Cervical	7	Neck region
Thoracic	12	Chest region; each vertebra articulates with a rib
Lumbar	5	Lower back
Sacrum	1	Fusion of 5 sacral vertebrae
Coccyx	1	Fusion of 4 coccygeal vertebrae

Functions:

- Protect spinal cord.
- Support head and trunk.
- Attachment for ribs and back muscles.

Houses red marrow in some regions



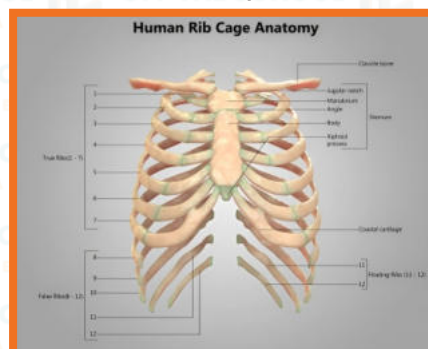
6. Thoracic Cage (25 Bones)

1. Ribs (24; 12 pairs)

- True ribs (1–7): attach directly to sternum via costal cartilage.
- False ribs (8–12): attach indirectly (8–10) or not at all (11–12) to sternum.
- Floating ribs (11–12): no anterior attachment.
- Function: Protect thoracic organs; support breathing movements.

2. Sternum (1)

- Manubrium, body, xiphoid process.
- Articulates with clavicles and first seven pairs of ribs.





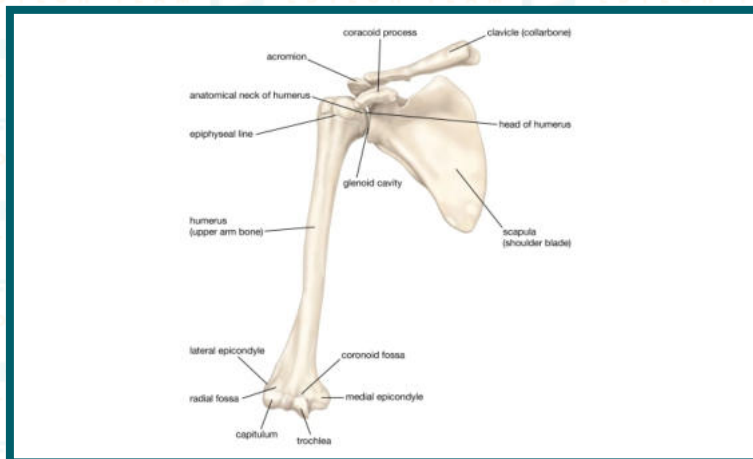
3. Appendicular Skeleton (126 Bones)

1. Overview

- Total Bones: **126**.
- **Divisions:**
 1. Pectoral (Shoulder) Girdle & Upper Limbs.
 2. Pelvic (Hip) Girdle & Lower Limbs.

2. Pectoral Girdle & Upper Limb Bones (64 Bones Total)

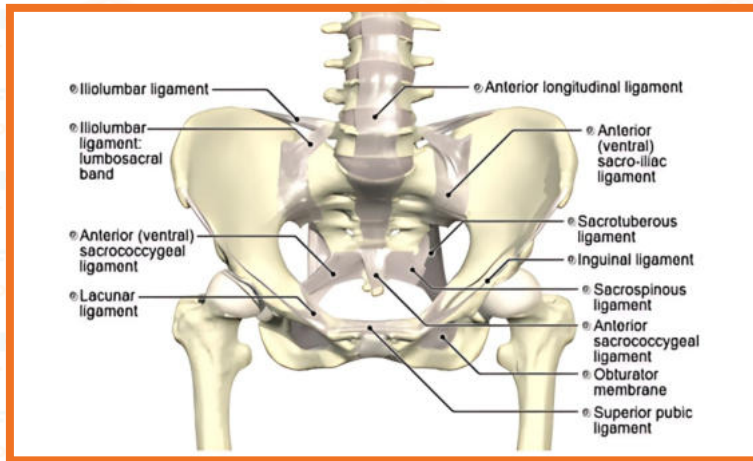
- Pectoral Girdle (4 bones).
- Clavicles: 2.
- Scapulae: 2
- **Function:** Anchors upper limbs to trunk; provides attachment for shoulder and arm muscles; allows wide range of motion.
- Upper Limbs (60 bones).
- Humerus: 2 (arm).
- Radius: 2 (forearm, lateral).
- Ulna: 2 (forearm, medial).
- Carpals: 16 (wrist; 8 per hand).
- Metacarpals: 10 (palm; 5 per hand).
- Phalanges: 28 (fingers; 14 per hand).



4. Pelvic Girdle & Lower Limb Bones (62 Bones Total)

- Pelvic Girdle (6 bones → 2 coxal bones).
- Ilium: 2.
- Pubis: 2.
- Ischium: 2.
- Fuses into: 2 coxal (hip) bones.
- Function: Supports body weight in standing and locomotion; protects pelvic organs (bladder, reproductive structures, lower intestines).
- Lower Limbs (56 bones).
- Femur: 2 (thigh).
- Patella: 2 (kneecap).
- Tibia: 2 (shin; medial).
- Fibula: 2 (lateral lower leg).
- Tarsals: 14 (ankle; 7 per foot).
- Metatarsals: 10 (foot arch; 5 per foot).
- Phalanges: 28 (toes; 14 per foot).





5. Key Functions & Features

- **Mobility & Leverage:** Limb bones act as levers; joints provide a wide range of movement.
- **Muscle Attachment:** Girdles and limb bones bear numerous tubercles, crests, and fossae for muscle and ligament attachment.
- **Weight Bearing:** Pelvic girdle transfers weight from axial skeleton to lower limbs; lower limbs support body weight during standing, walking, and running.
- **Fine Manipulation:** Upper limbs (especially the hand) enable precise, dexterous movements.

Types of Joints

1. Definition

- **Joint (Articulation):** Functional junction where two or more bones meet
- **Approximate Count in Human Body:** 360

2. Classification by Tissue & Mobility

1. Fibrous Joints (Immovable to slightly movable).
2. Cartilaginous Joints (Slightly movable).
3. Synovial Joints (Freely movable).

3. Fibrous Joints

- **Binding Tissue:** Dense connective tissue rich in collagen.
- **Subtypes & Examples:**
 1. Sutures (skull bones; immovable).
 2. Syndesmoses (interosseous membrane between radius & ulna; slight movement).
 3. Gomphoses (tooth in alveolar socket; immovable).

4. Cartilaginous Joints

- **Binding Tissue:** Hyaline cartilage or fibrocartilage.
- **Subtypes & Examples:**

1. Synchondroses

- Hyaline cartilage plate.
- e.g., epiphyseal (growth) plate; first rib to sternum.

2. Symphyses

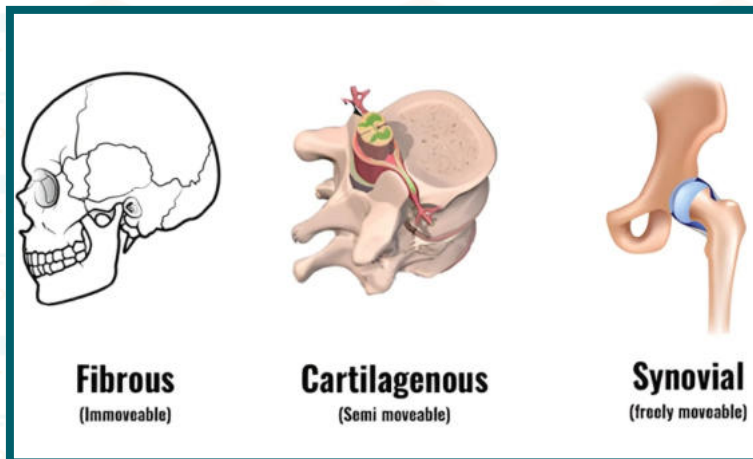
- Fibrocartilage pad between bones.
- e.g., intervertebral discs; pubic symphysis.



6. Synovial Joints

Key Features:

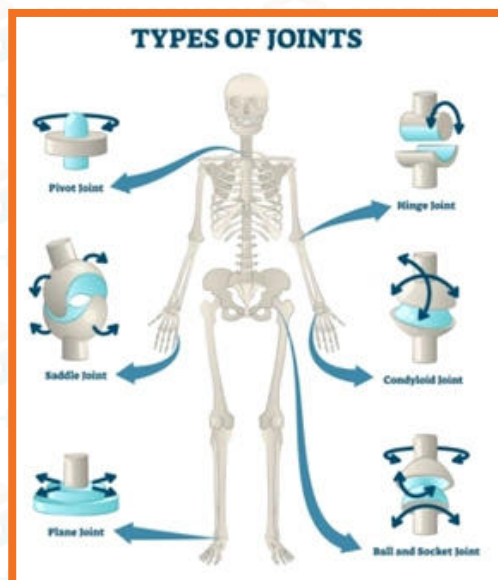
- **Joint Capsule:** Fibrous outer layer + inner synovial membrane.
- **Synovial Cavity:** Space filled with synovial fluid.
- **Synovial Fluid:** Lubricates & nourishes articular cartilage.
- **Articular Cartilage:** Hyaline cartilage covering bone ends.
- **Supporting Structures:** Ligaments, bursae, menisci or articular discs.
- **Mobility:** Freely movable; classified by movement type.



Types of Synovial Joints

Joint Type	Description	Example
Hinge	Uniaxial flexion/extension	Elbow (humerus-ulna), knee
Pivot	Uniaxial rotation around longitudinal axis	Proximal radio-ulnar joint
Ball & Socket	Multiaxial; flexion/extension, abduction/adduction, rotation	Shoulder (glenohumeral), hip
Condylloid (Ellipsoid)	Biaxial; flexion/extension & ab-adduction	Wrist (radiocarpal), metacarpophalangeal
Saddle	Biaxial; similar to condylloid but greater range	Carpometacarpal of thumb
Plane (Gliding)	Nonaxial; sliding or gliding movements	Intercarpal, intertarsal, vertebral articular processes





Disorders of the Skeleton

1. Overview

- **Definition:** Conditions that impair the structure or function of bones, joints, and associated connective tissues.

- **Common Disorders:**

- Spondylosis.
- Sciatica.
- Disc slip (herniated disc).
- Arthritis.

2. Spondylosis

- **Definition:** Degenerative changes in vertebrae and intervertebral discs.

- **Pathology:**

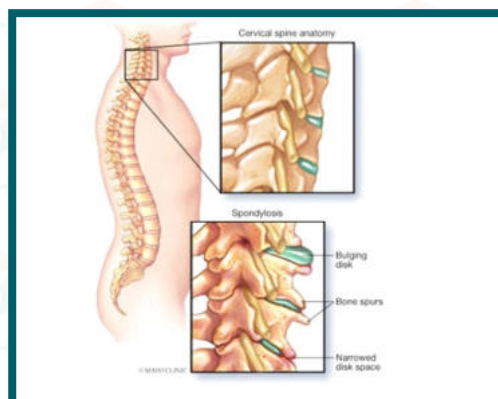
- Formation of bone spurs (osteophytes).
- Narrowing of intervertebral foramina and disc space.
- Disc degeneration from aging.

- **Locations:**

- Cervical spine (neck).
- Thoracic spine (upper/mid back).
- Lumbar spine (lower back) – most common.

- **Symptoms:**

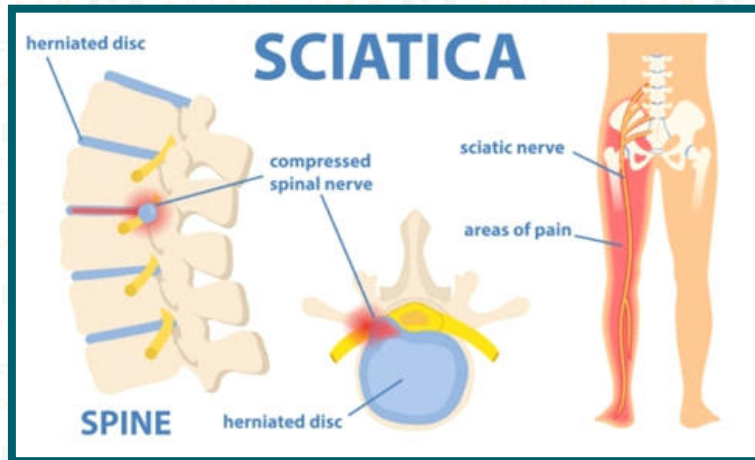
- Localized back or neck pain.
- Radiating pain, numbness, tingling from nerve compression.





3. Sciatica

- **Definition:** Pain resulting from compression or irritation of the sciatic nerve.
- **Anatomy:**
- Formed by nerve roots L4–S3.
- Exits pelvis through sciatic notch, runs down posterior thigh, branches below knee.
- **Etiology:**
- Herniated disc.
- Spinal stenosis.
- Piriformis syndrome.
- **Symptoms:**
- Sharp, shooting pain from lower back into buttock and down leg.
- Numbness, tingling, muscle weakness in affected leg.
- Onset may be sudden or gradual.



4. Disc Slip (Herniated/Prolapsed Disc)

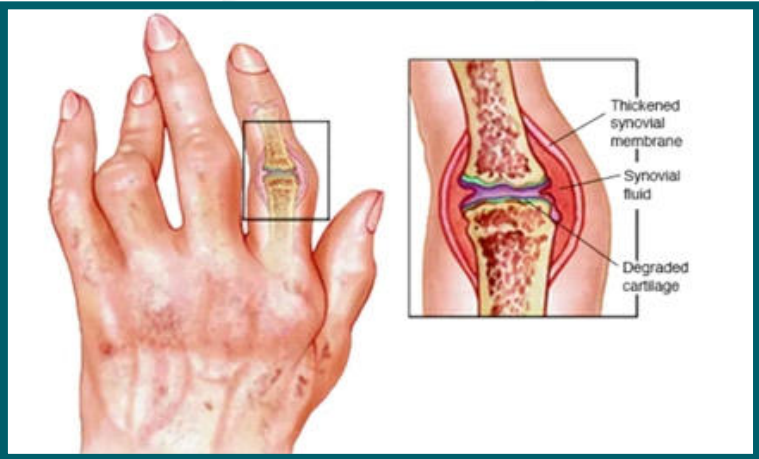
- **Definition:** Protrusion of nucleus pulposus through a tear in annulus fibrosus.
- **Intervertebral Disc Structure:**
- **Annulus fibrosus:** Tough outer ring.
- **Nucleus pulposus:** Soft, gelatinous core.
- **Pathophysiology:**
- Injury or degeneration weakens annulus fibrosus.
- Nucleus pulposus bulges or ruptures into spinal canal.
- **Symptoms:**
- Severe, localized back pain.
- Radicular pain if nerve roots are compressed.
- Numbness or pain radiating into limbs.



5.Arthritis

- **Definition:** Inflammation or degeneration of one or more joints.
- **General Symptoms:**
- Joint pain and stiffness.
- Swelling, reduced range of motion.
- Functional impairment.

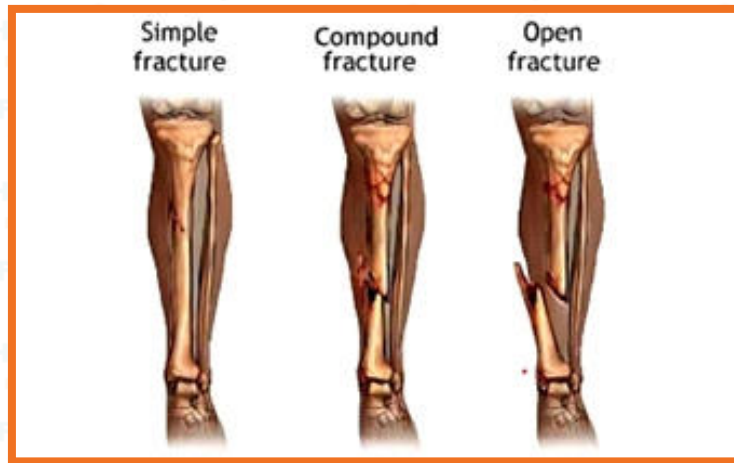
Type	Pathology	Key Features & Examples
Osteoarthritis (OA)	Degeneration of articular cartilage	"Wear-and-tear"; e.g., weight-bearing joints (knees, hips)
Rheumatoid Arthritis (RA)	Autoimmune attack on synovial membrane	Symmetric joint involvement; systemic inflammation
Ankylosing Spondy	Chronic inflammation of spine & sacroiliac joints	Progressive spinal rigidity; primarily in lower back
litisGout	Deposition of monosodium urate crystals in joints	Acute, painful flares; commonly affects first metatarsophalangeal joint ("big toe")



Types of Bone Fractures & Repair

1.Fracture Classification

Type	Description
Simple (Closed)	Bone cracks or breaks without piercing the skin; no external wound
Compound (Open)	Broken bone protrudes through the skin, creating an open wound
Complicated	Fracture with additional damage to surrounding tissues (muscles, blood vessels, nerves, organs)



2. Common Signs & Symptoms

- Pain at the injury site, often severe.
- Tenderness to touch.
- Crepitus ("grating" or "cracking" sensation with movement).
- Deformity or abnormal shape of the limb/body part.
- Swelling and bruising around the fracture.
- Inability to bear weight or use the affected area.
- Abnormal mobility (movement in places where bone should be rigid).

3. Four Stages of Simple Fracture Repair

a) Hematoma Formation

- **Timeframe:** First few days post-fracture.
- **Process:**
 - Blood vessels in bone and periosteum rupture → hematoma (clot) forms.
 - Hematoma seals fracture site, prevents further bleeding.
 - Inflammatory cells clear debris; pain and swelling occur.

b) Fibrocartilaginous (Soft) Callus Formation

- **Timeframe:** 1–3 weeks post-fracture.
- **Process:**
 - New blood vessels invade hematoma.
 - Osteogenic cells (from periosteum/endosteum) differentiate into osteoblasts and chondroblasts.
 - Granulation tissue forms, creating a collagen-fibrin mesh.
 - Fibrocartilage ("soft callus") bridges fracture gap.

c) Bony (Hard) Callus Formation

- **Timeframe:** 3 weeks to 3 months post-fracture.
- **Process:**
 - Soft callus is gradually replaced by woven bone as osteoblasts deposit mineralized matrix.
 - A "hard callus" of spongy bone unites fracture fragments.
 - Osteoclasts and osteoblasts work together to remodel this immature bone.

d) Bone Remodeling

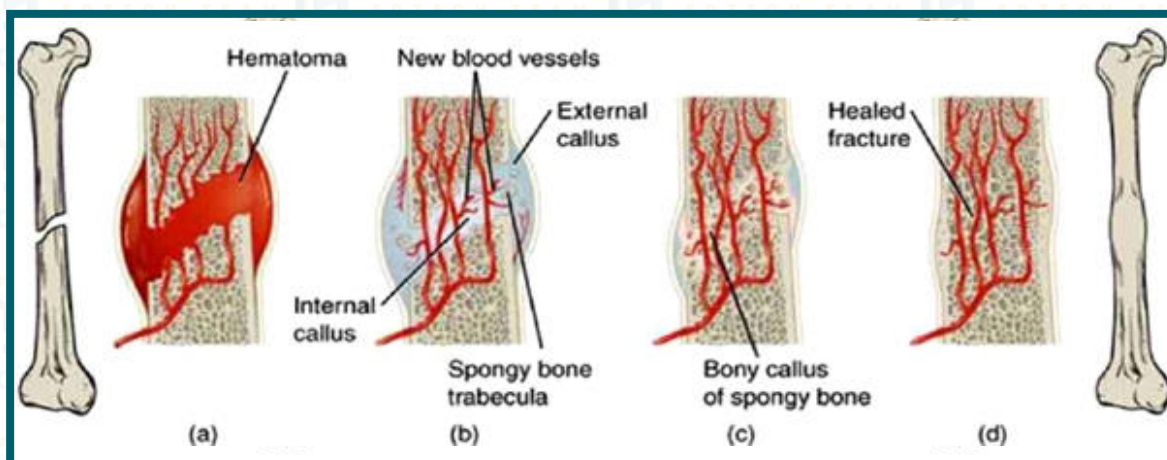
- **Timeframe:** Several months to years.





Process:

- Osteoclasts resorb excess callus and damaged bone.
- Osteoblasts fill in resorption cavities with new lamellar bone.
- Compact bone restores original shape, strength, and internal structure.
- Final result approximates pre-injury bone architecture.



Joint Injuries

1. Overview

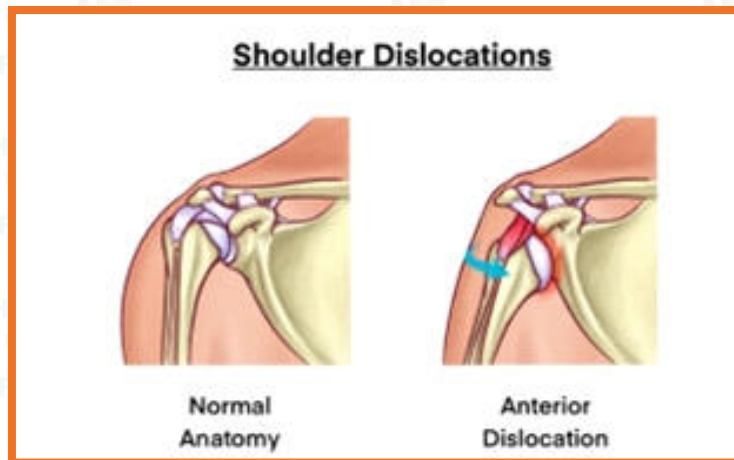
- **Definition:** Damage to a joint's supporting structures (ligaments, capsule, surrounding tissues).
- **Common Causes:**
 - Trauma or sudden impact.
 - Twisting or overextension.
 - Repetitive stress or heavy use.
 - Underlying joint instability (e.g., arthritis, muscle weakness).

2. Dislocations

- **Definition:** Complete separation or abnormal displacement of the articulating bone surfaces in a joint.
- **Common Sites:** Shoulder (glenohumeral), elbow, hip, finger joints.
- **Mechanism:**
 - High-force trauma or fall.
 - Excessive twisting beyond normal range.
- **Types (shoulder example):**
 - Anterior dislocation (most common; humeral head displaced forward).
 - Posterior dislocation (less common; humeral head displaced backward).
- **Key Symptoms:**
 - Intense, sudden pain.
 - Visible deformity or "out-of-place" appearance.
 - Swelling and bruising around joint.
 - Inability to move or bear weight on the joint.
 - Sensation of joint instability or "giving way".
- **Complications:**
 - Nerve or blood-vessel injury.
 - Recurrent dislocations if ligaments remain lax.



- **Management:**
- Immediate care: Immobilize, ice, pain control.
- Reduction: Closed (manual) or open (surgical) realignment.
- Rehabilitation: Rest, protected mobilization, strengthening exercises.
- Surgery: In cases of recurrent instability or associated fractures.



3. Sprains

- **Definition:** Stretching or tearing of ligaments—the fibrous bands connecting bone to bone.

- **Common Sites:** Ankle (lateral ligaments), knee (ACL, MCL), wrist.

- **Grading:**

- **Grade I (Mild):** Microscopic tears, mild tenderness/swelling, no instability.

- **Grade II (Moderate):** Partial ligament tear, moderate pain/swelling, some joint laxity.

- **Grade III (Severe):** Complete ligament rupture, severe pain/swelling, significant instability.

- **Mechanism:**

- Twisting or rolling of the limb (e.g., inversion of the ankle).

- Direct blow to the joint.

- **Types of Ankle Sprains:**

- **Inversion sprain:** Lateral ligament complex injured by inward roll.

- **Eversion sprain:** Medial (deltoid) ligament injured by outward roll.

- **Key Symptoms:**

- Pain localized over ligament.

- Swelling and bruising within hours.

- Tenderness on palpation of ligament.

- Difficulty bearing weight or moving the joint.

- **Management (R.I.C.E. & beyond):**

- **Rest:** Avoid activities that stress the ligament.

- **Ice:** 15–20 min every 2–3 h for 48 h.

- **Compression:** Elastic bandage to limit swelling.

- **Elevation:** Above heart level to reduce edema.

- **Immobilization:** Brace or splint for Grade II/III.

- **Rehabilitation:** Gradual range-of-motion and strengthening exercises.

- **Surgery:** Rare, for complete ruptures or chronic instability.





Muscles

1. Definition & General Functions

- **Muscle:** Bundle of contractile tissue capable of shortening (contraction) and lengthening (relaxation) to produce movement.
- **Primary Roles:**
 - Generate body movement (locomotion, posture).
 - Propel body fluids (e.g., lymph, blood).
 - Produce heat via metabolic activity.

2. Muscle Types

Feature	Smooth Muscle	Cardiac Muscle	Skeletal Muscle
Location	Walls of hollow organs (e.g., GI tract, blood vessels, bladder, uterus)	Heart wall (myocardium)	Attached to bones via tendons
Control	Involuntary (autonomic nervous system & hormones)	Involuntary (intrinsic pacemaker & autonomic input)	Voluntary (somatic nervous system)
Appearance	Non-striated; spindle-shaped cells with single nucleus	Striated; branched fibers; 1–2 nuclei; intercalated discs	Striated; long cylindrical fibers; multinucleated
Special Features	Slow, sustained contractions; resistant to fatigue	Syncytium via intercalated discs for rapid impulse spread; high mitochondrial density	Fast, forceful contractions; great regenerative capacity via satellite cells
Function	Regulate flow/pressure in vessels; move contents through organ systems	Pump blood continuously; coordinate heartbeat	Generate body and limb movements; maintain posture; generate heat



3. Detailed Characteristics

Smooth Muscle:

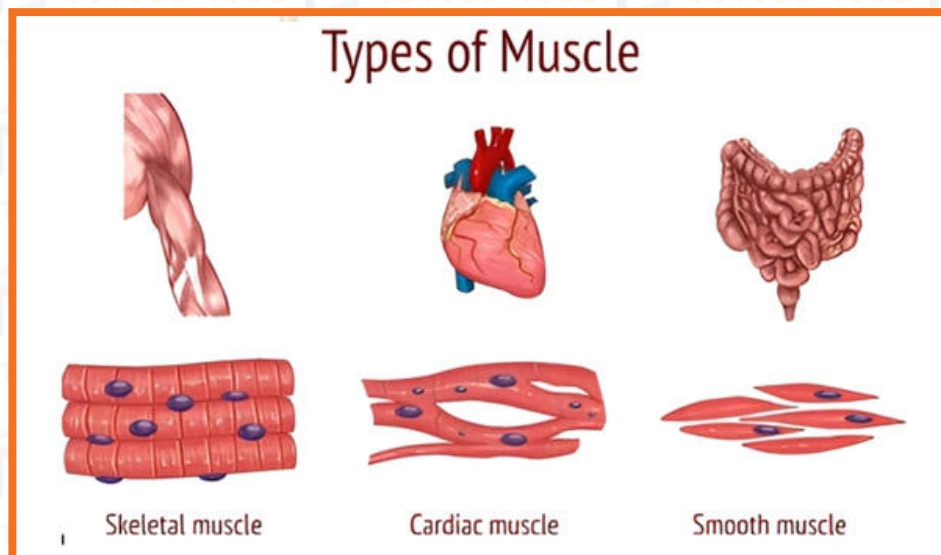
- **Cells:** Spindle-shaped (tapered ends), single central nucleus.
- **Contraction:** Slow and sustained; can maintain tone with low energy.
- **Regulation:** Influenced by stretch, autonomic nerves, and circulating hormones.

Cardiac Muscle:

- **Cells:** Short, branching fibers; 1–2 centrally located nuclei.
- **Intercalated Discs:**
 - Contain gap junctions for electrical coupling.
 - Contain desmosomes for mechanical strength.
- **Syncytium:** Functional syncytium allows rapid, coordinated contraction.
- **Metabolism:** Rich in mitochondria; continuous oxidative metabolism prevents fatigue.

Skeletal Muscle:

- **Cells (Fibers):** Very long, cylindrical; multiple peripheral nuclei.
- **Striations:** Alternating A-bands and I-bands visible microscopically.
- **Motor Units:** One motor neuron plus all fibers it innervates; basis for gradation of force.
- **Regeneration:** Satellite cells (muscle stem cells) enable limited repair.

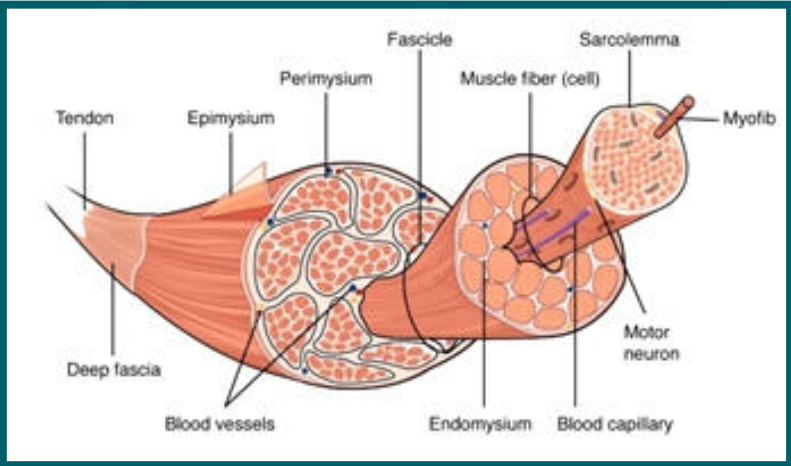


Structure of Skeletal Muscle

1. Organization of the Muscle Organ

- **Muscle:**
 - Entire muscle is surrounded by **epimysium** (dense irregular connective tissue).
 - Contains bundles (fascicles) of muscle fibers.
- **Fascicle:**
 - A discrete bundle of muscle fibers.
 - Enclosed by **perimysium** (connective tissue sheath).
- **Muscle Fiber (Cell):**
 - Long, cylindrical, multinucleated cell.
 - Wrapped in **endomysium** (fine reticular collagen).





Layer	Encloses	Composition & Function
Epimysium	Entire muscle	Dense irregular CT; provides overall protection; allows force transmission
Perimysium	Each fascicle	Collagenous CT; houses blood vessels and nerves; divides muscle into units
Endomysium	Individual muscle fiber	Reticular CT; supports capillaries and nerve fibers; binds fibers together

3. Neurovascular Supply

- **Blood Vessels:**
- Highly vascularized: each fascicle and fiber receives capillary networks.
- Supplies O₂, nutrients; removes metabolic waste.
- **Nerves:**
- Motor neurons branch to innervate individual muscle fibers.
- Neuromuscular junctions transmit action potentials to trigger contraction.

Ultrastructure of a Skeletal Muscle Fiber

1. General Cell Features

- **Muscle Fiber (Myocyte):**
- Long, cylindrical, multinucleated cell.
- Cytoplasm = **sarcoplasm** (contains nuclei, abundant mitochondria, glycogen).
- Plasma membrane = **sarcolemma**.

2. Neuromuscular Junction

- **Motor End Plate:** Region of sarcolemma folded into junctional folds.
- **Axon Terminal:** Synaptic vesicles release acetylcholine (ACh).
- **Synaptic Cleft:** Space between axon terminal and sarcolemma.
- **Function:** ACh binds receptors on sarcolemma → triggers action potential.



3. Membranous Networks

- **Sarcoplasmic Reticulum (SR):**
 - Specialized smooth ER surrounding each myofibril.
- **Terminal cisternae:** Enlarged SR regions adjacent to T-tubules.
- **Function:** Stores and releases Ca^{2+} for contraction.
- **T-Tubules (Transverse Tubules):**
 - Invaginations of sarcolemma that penetrate cell interior.
 - Form "triads" where one T-tubule is flanked by two SR terminal cisternae.
- **Function:** Rapidly conduct action potentials to SR.

4. Myofibrils & Contractile Proteins

- **Myofibrils:** Bundles of protein filaments filling the sarcoplasm.
- **Filament Types:**
 - **Thick filaments:** Myosin (≈ 16 nm diameter).
 - Two heavy chains twisted with globular heads (cross-bridge sites).
 - **Thin filaments:** Actin ($\approx 7-8$ nm diameter).
 - Actin monomers with binding sites for myosin.
 - Associated regulatory proteins:..
- **Tropomyosin:** Blocks binding sites at rest.
- **Troponin complex:**
 - **Tn-C:** Binds Ca^{2+} .
 - **Tn-I:** Inhibits actomyosin interaction.
 - **Tn-T:** Anchors troponin to tropomyosin.

5. Striated Banding Pattern

- **Sarcomere:** Functional contractile unit between two Z-lines.
- **Bands & Zones:**
 - **I-band:** Light region containing only thin (actin) filaments; bisected by Z-line.
 - **A-band:** Dark region containing overlapping thick (myosin) and thin filaments.
 - **H-zone:** Central lighter region of A-band with only thick filaments.
 - **M-line:** Midpoint of H-zone; proteins that anchor thick filaments.

6. Contraction Essentials

- **Ca^{2+} Release:** Action potential $\rightarrow$ SR releases Ca^{2+} into sarcoplasm.
- **Cross-Bridge Cycling:** Ca^{2+} binds Tn-C $\rightarrow$ tropomyosin shifts $\rightarrow$ myosin heads bind actin $\rightarrow$ power stroke.

Sarcomere of Skeletal Muscle

1. Definition

- **Sarcomere:** The basic contractile unit of a myofibril, extending from one Z-line to the next.

2. Major Regions & Lines

Region/Line	Composition	Function / Note
Z-line (Z-disc)	A network of proteins (α -actinin) anchoring thin filaments	Defines sarcomere boundaries; attachment site for actin
I-band	Light band containing only thin (actin) filaments	Shortens during contraction; bisected by Z-line

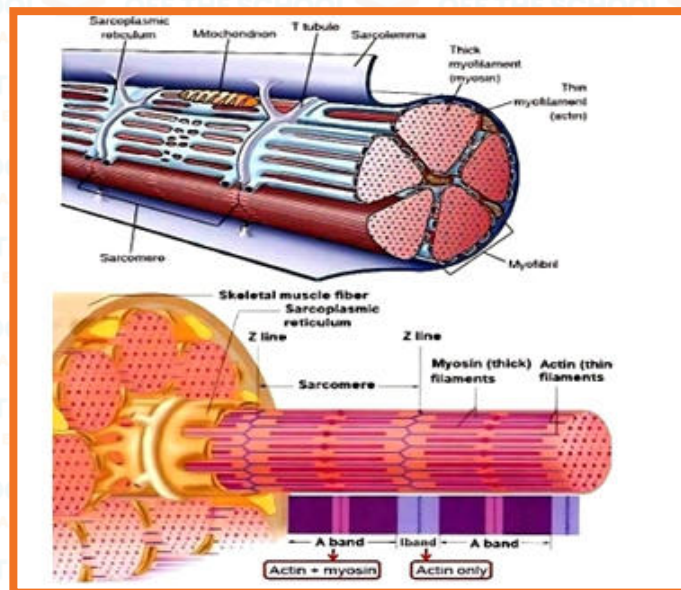




Region/Line	Composition	Function / Note
A-band	Dark band where thick (myosin) filaments overlap thin filaments	Remains constant length during contraction
H-zone	Lighter central region of A-band containing only thick filaments	Disappears as thin filaments slide into center during contraction
M-line	Protein myomesin cross-links adjacent thick filaments	Stabilizes myosin filament arrangement at sarcomere center

3. Filament Organization

- **Thick Filaments (Myosin):**
 - Composed of myosin molecules with globular heads (cross-bridges) projecting outward
 - Heads bind to actin and hydrolyze ATP to generate force.
- **Thin Filaments (Actin):**
 - Double-helix of actin monomers, each with a myosin-binding site.
 - Associated with tropomyosin (blocks binding sites) and troponin complex (regulates binding).



Sliding Filament Model of Muscle Contraction

1. Overview & Key Principle

- **Originators:** Hugh Huxley & Jean Hanson (1954).
- **Core Concept:** During contraction, thin (actin) and thick (myosin) filaments slide past each other without changing their individual lengths.
- **Sarcomere Changes:**
 - Z-lines move closer together.
 - I-bands and H-zones narrow.
 - A-band (filament overlap region) remains constant in length.





2. Cross-Bridge Cycle (Contraction Sub-Stages)

1. Neuromuscular Stimulation & Ca^{2+} Release:

- Motor neuron fires $\rightarrow$ acetylcholine released at motor end plate.
- Sarcolemma/T-tubule depolarization opens Ca^{2+} channels in sarcoplasmic reticulum.
- Ca^{2+} floods into sarcoplasm and binds troponin C on thin filaments.

2. Exposure of Myosin-Binding Sites:

- Ca^{2+} -troponin interaction shifts tropomyosin off actin's binding sites.
- Actin's myosin-binding sites become available.

3. Cross-Bridge Formation:

- Myosin heads, energized by ATP hydrolysis ($\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ via myosin ATPase), cock into a high-energy position.
- Myosin heads bind to exposed sites on actin, forming "**cross-bridges**".

4. Power Stroke:

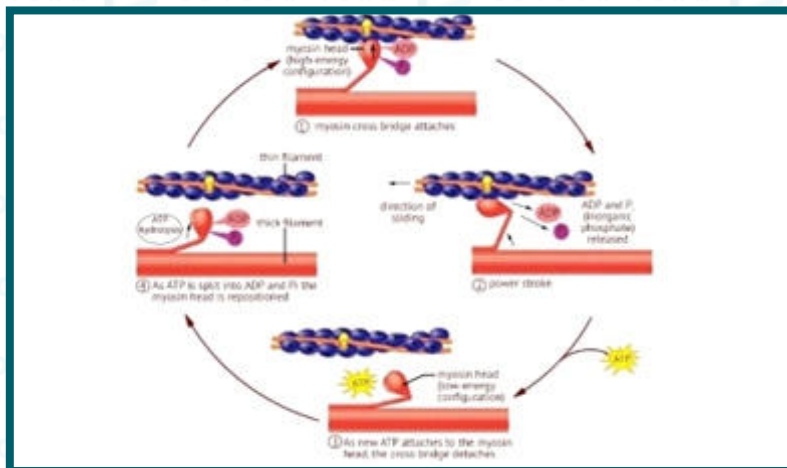
- Release of $\text{ADP} + \text{P}_i$ triggers myosin head pivot toward M-line.
- Actin filaments are pulled inward, shortening the sarcomere.

5. Cross-Bridge Detachment:

- Binding of a new ATP molecule to myosin head causes detachment from actin.

6. Re-cocking of Myosin Head:

- ATP hydrolysis re-energizes (recocks) myosin head, readying it for another cycle.
- **Cycle Repeats:** as long as Ca^{2+} and ATP are present.



Relaxation Mechanisms

1. Termination of Neural Signal

- Acetylcholine in synaptic cleft is rapidly broken down by acetylcholinesterase.
- Cessation of action potentials in sarcolemma/T-tubules.

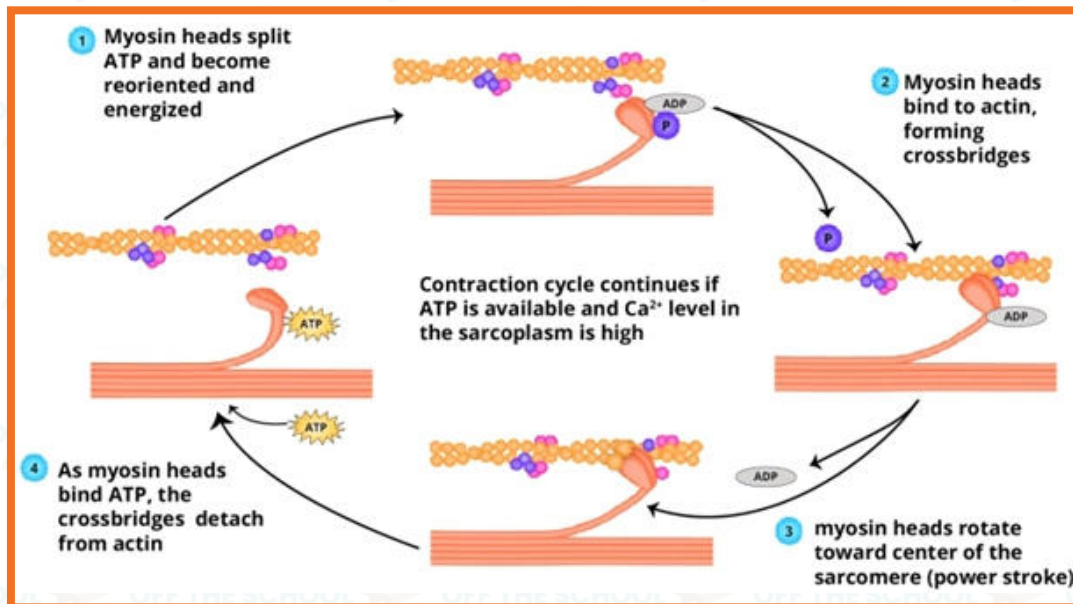
2. Ca^{2+} Re-uptake

- ATP-driven Ca^{2+} pumps (SERCA) transport Ca^{2+} back into sarcoplasmic reticulum.
- Reduced cytosolic Ca^{2+} causes troponin-tropomyosin complex to block actin's binding sites.

3. Return to Resting State

- Cross-bridge cycling stops.
- Sarcomere returns to resting length under passive tension.





Antagonistic Muscle Action at the Knee Joint

1. Antagonism & Agonism

- **Antagonism:** Ability of one muscle to oppose the action of another.
- **Agonist (Prime Mover):** Muscle that contracts to produce a desired movement.
- **Antagonist:** Muscle that relaxes while the agonist contracts, then may contract to return the joint to its original position.

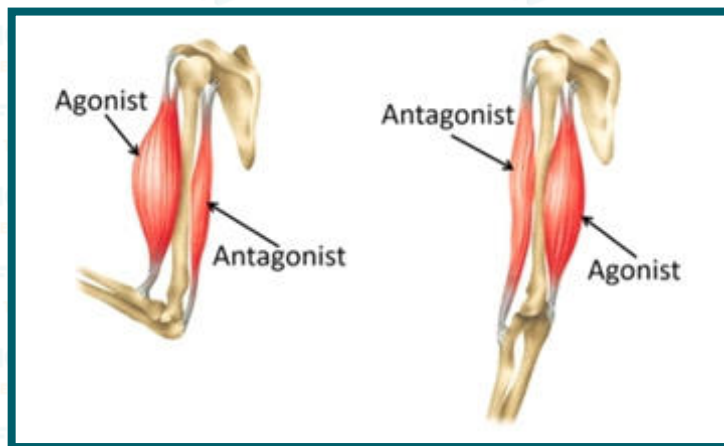
2. Knee Joint Anatomy

- **Type:** Synovial hinge joint (with limited rotation).
- **Articulations:**
 - Distal femur (medial & lateral condyles) $\leftrightarrow$ Proximal tibia (tibial plateau).
 - Patella articulates with femoral patellar surface.
- **Primary Movements:**
 - **Flexion:** Decreases angle between thigh and shin.
 - **Extension:** Increases angle to straighten the leg.

3. Agonist–Antagonist Pairs at the Knee

Movement	Agonist (Contracts)	Antagonist (Relaxes)
Flexion	Hamstring group (biceps femoris, semitendinosus, semimembranosus)	Quadriceps group
Extension	Quadriceps group (rectus femoris, vastus lateralis, vastus medialis, vastus intermedius)	Hamstring group





4. Hamstring Muscles (Flexors)

Components:

- **Biceps femoris** (long & short head)

- **Semitendinosus**

- **Semimembranosus**

- **Origin:** Ischial tuberosity (plus femur for short head of biceps femoris)

Insertion:

- Biceps → head of fibula & lateral tibia

- Semis → medial tibia

Actions:

- Knee flexion

- Hip extension (long head)

- Tibial rotation (medial with semis; lateral with biceps)

Clinical Note:

- Hamstring strain: Overstretching can cause micro-tears, ranging from mild (Grade I) to severe (Grade III)

5. Quadriceps Muscles (Extensors)

Components:

- Rectus femoris (also hip flexor)

- Vastus lateralis, vastus medialis, vastus intermedius

Origin:

- Rectus → anterior inferior iliac spine

- Vastus muscles → various regions of femoral shaft

- **Insertion:** Patellar tendon → tibial tuberosity

Actions:

- Knee extension (primary)

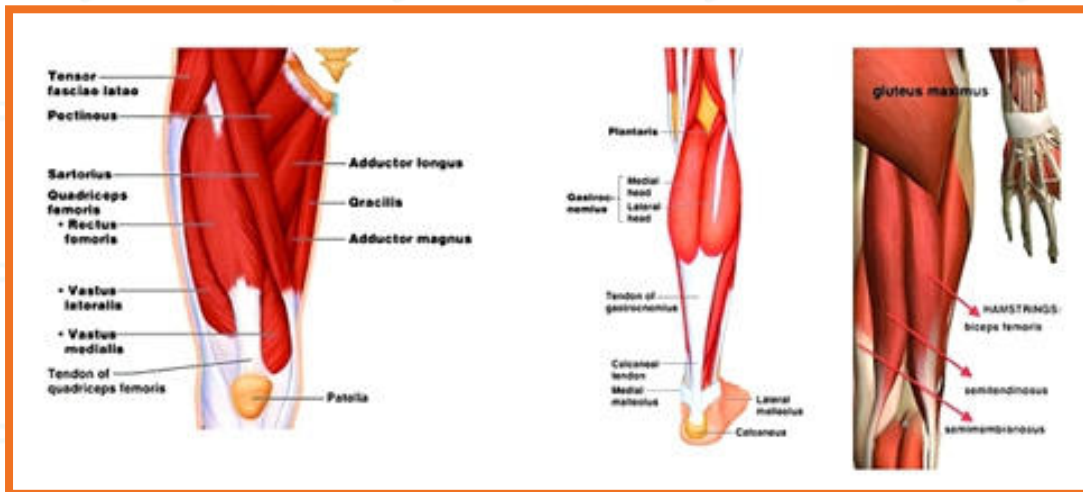
- Hip flexion (rectus femoris)

Function in Activities:

- Kicking, jumping, cycling, running

- Stabilizing knee during standing



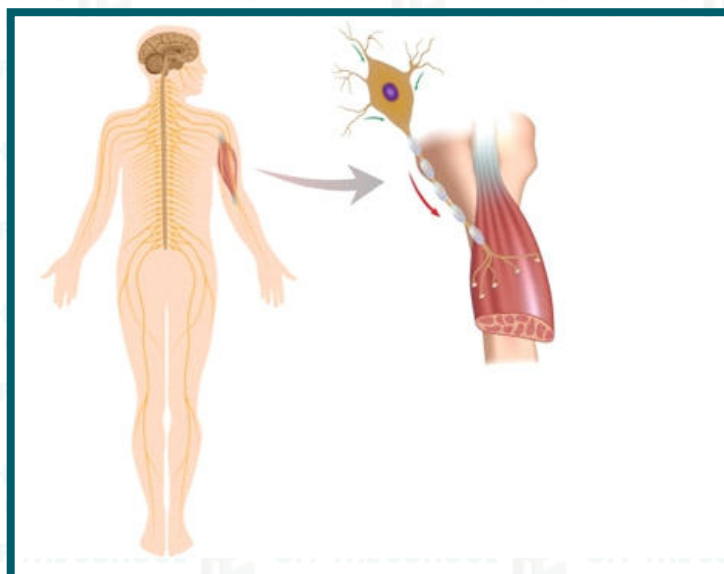


6. Reciprocal Innervation & Coordination

- When the **hamstrings** contract to flex the knee, the **quadriceps** are neurologically inhibited (relaxed) to allow smooth movement.
- Conversely, during knee extension, the quadriceps contract while hamstrings are inhibited.

1. Muscle Fatigue:

- **Definition:** Transient decline in a muscle's ability to generate force despite continued stimulation.
- **Primary Cause:** Accumulation of lactic acid from anaerobic ATP production.
- **Mechanism:**
 - Intense exercise $\rightarrow$ O_2 demand exceeds supply $\rightarrow$ anaerobic glycolysis $\uparrow$ $\rightarrow$ lactic acid builds up
 - $\downarrow$ pH in sarcoplasm inhibits cross-bridge cycling $\rightarrow$ muscle fibers less responsive to Ca^{2+} signals
- **Recovery:**
 - Lactic acid diffuses into bloodstream $\rightarrow$ transported to liver (Cori cycle) $\rightarrow$ converted to glucose (requires ATP and O_2)
 - Rest and adequate oxygenation restore pH and ATP levels $\rightarrow$ muscle function returns





2. Muscle Cramps

● **Definition:** Sudden, involuntary, and often painful contraction of a muscle or muscle group.

● **Common Triggers:**

- Overuse or prolonged contraction during exercise
- Dehydration or electrolyte imbalances ($\downarrow \text{Ca}^{2+}, \text{K}^{+}, \text{Na}^{+}, \text{Mg}^{2+}$)
- Inadequate blood flow (e.g., peripheral arterial disease)

● **Symptoms:**

- Sharp pain, palpable tightness, visible bulging under skin
- May last seconds to several minutes

● **Management:**

- **Immediate:** Gentle stretching of the affected muscle
- **Supportive:** Massage, heat application for relaxation, cold for soreness

● **Prevention:**

- Maintain hydration and electrolyte balance
- Gradual warm-up and cool-down routines during exercise



3. Tetany vs. Tetanus

Feature	Tetany	Tetanus
Definition	Involuntary, prolonged muscle contractions	Infectious disease causing muscle spasms & rigidity
Etiology	Electrolyte imbalances ($\downarrow \text{Ca}^{2+}, \text{Mg}^{2+}$; acid-base disturbances)	Clostridium tetani infection → tetanospasmin toxin
Pathophysiology	Increased nerve/muscle excitability	Toxin blocks inhibitory neurotransmitters (Glycine, GABA) at spinal synapses
Clinical Features	Carpopedal spasms, facial "tetany" (Chvostek's, Trousseau's signs)	Lockjaw, opisthotonus, generalized "risus sardonicus"





Feature	Tetany	Tetanus
Onset	May be acute (rapid Ca^{2+} drop) or chronic	3–21 days post-wound (depends on inoculum)
Treatment	Correct electrolytes; treat underlying metabolic imbalance	Wound debridement; human tetanus immunoglobulin; antibiotics; muscle relaxants; supportive ICU care
Prognosis	Excellent with prompt electrolyte correction	Variable; high mortality without timely treatment and ventilation support

